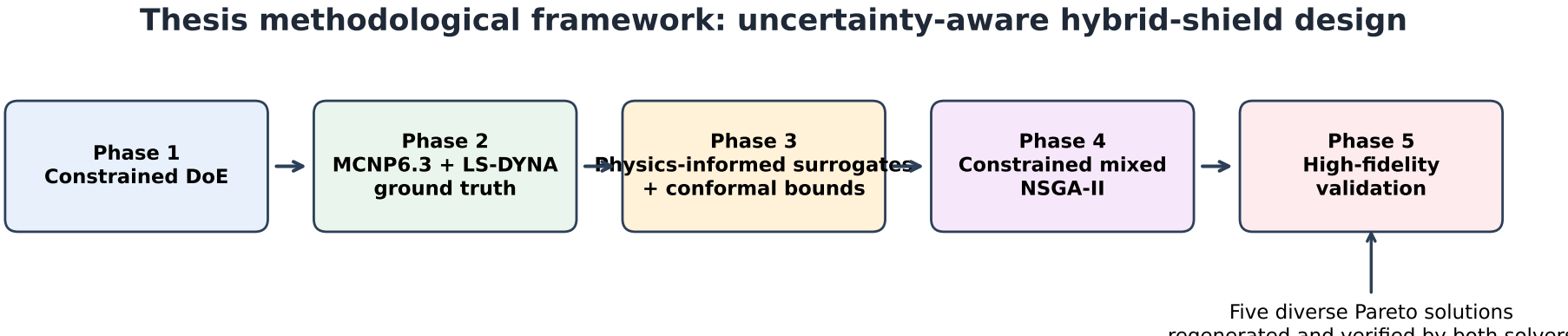
Thesis methodological framework: uncertainty-aware hybrid-shield design
Phase 1
Constrained DoE
Phase 2
MCNP6.3 + LS-DYNA
ground truth
Phase 3
Physics-informed surrogates
+ conformal bounds
Phase 4
Constrained mixed
NSGA-II
Phase 5
High-fidelity
validation
Five diverse Pareto solutions
regenerated and verified by both solvers